

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Short Answer Text(High)
Assessment Tool Weightage: 40%

QUESTIONS		
Q. No	Question	Bloom's Level
1	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember
2	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand
3	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand
4	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand
5	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember

Code of Conduct:

I MONIKA.B (192571002) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q1. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.

Schema :

A **schema** is the overall design or structure of a database. It defines how data is organized into tables, columns, relationships, and constraints. A schema is created once and changes very rarely.

Example :

A Student database contains:

Student(Student_ID, Name, Department, Age)

This table structure is called the **schema**.

Instance :

An **instance** is the actual data stored in the database at a particular time. It changes whenever records are inserted, updated, or deleted.

Example:

Student_ID	Name	Department	AGE	City
101	Monika	CSE-BS	20	Chennai
102	Rahul	AI	21	Coimbatore

Difference between Schema and Instance :

Schema	Instance
Structure of the database	Actual data stored
Changes rarely	Changes frequently
Defined during database design	Changes with database operations
Also called Intension	Also called Extension

Types of Schema :

1. Logical Schema :

- Describes the logical structure of the database.
- Defines tables, attributes, relationships, and constraints.
- Does not describe how data is physically stored.

Example:

Student(Student_ID, Name, Department)

2. Physical Schema :

- Describes how data is actually stored in memory or disk.
- Includes storage methods, indexing, and file organization.
- Managed by the DBMS.

3. View Schema :

- Defines what part of the database a particular user can see.
- Used for security and simplicity.

Example:

A faculty member may see only:

Student(Name, Department)

without seeing marks or personal details.

Conclusion :

Schema defines the structure of a database, while an instance represents the actual data. Logical, physical, and view schemas together form the three-schema architecture of a DBMS.

Q2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.**Entities and Attributes****1. Customer**

- Customer_ID (Primary Key)
- Name
- Address
- Phone_Number

- Email

2. Account

- Account_Number (Primary Key)
- Account_Type
- Balance
- Opening_Date

3. Branch

- Branch_ID (Primary Key)
- Branch_Name
- Location

4. Employee

- Employee_ID (Primary Key)
- Name
- Designation
- Salary

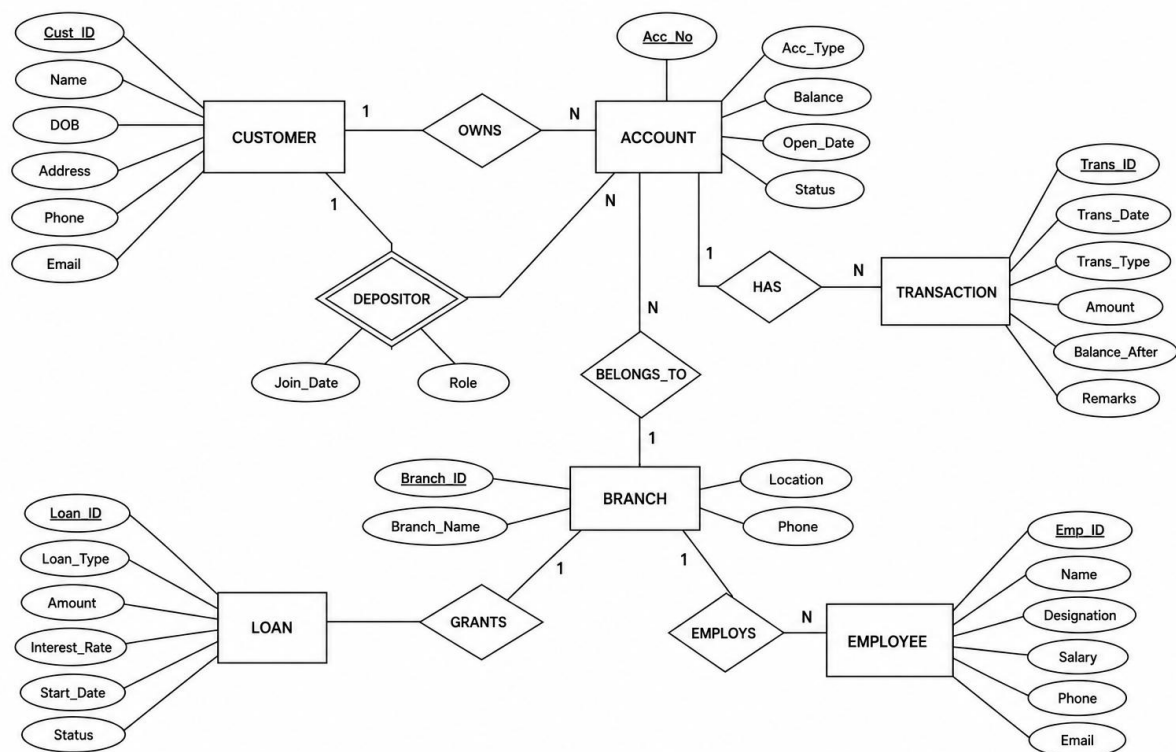
5. Loan

- Loan_ID (Primary Key)
- Loan_Type
- Amount
- Interest_Rate

Relationships :

- Customer **owns** Account (One customer can have many accounts.)
- Account **belongs to** Branch.
- Branch **employs** Employees.
- Customer **takes** Loan.
- Branch **sanctions** Loan.

ER Diagram :



Conclusion :

The ER diagram shows the entities, their attributes, and the relationships in a Bank Management System, making database design easier and more organized.

Q3. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.

Extended ER (EER) Model :

The **Extended Entity Relationship (EER) Model** is an advanced version of the ER model. It provides additional features to represent complex real-world situations more accurately.

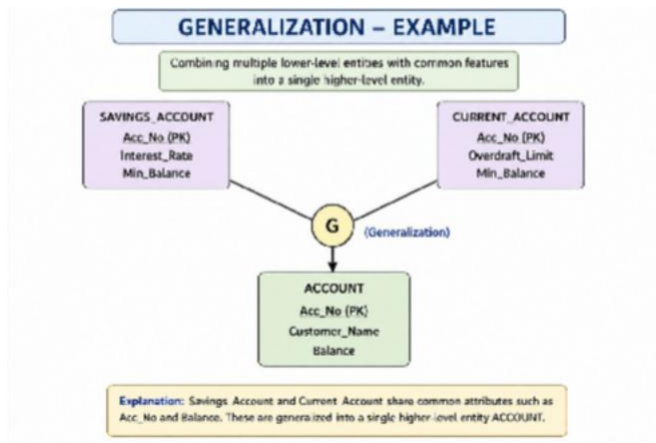
It includes:

- Generalization
- Specialization
- Aggregation
- Inheritance

1. Generalization :

Generalization combines similar lower-level entities into one higher-level entity based on common features.

Example

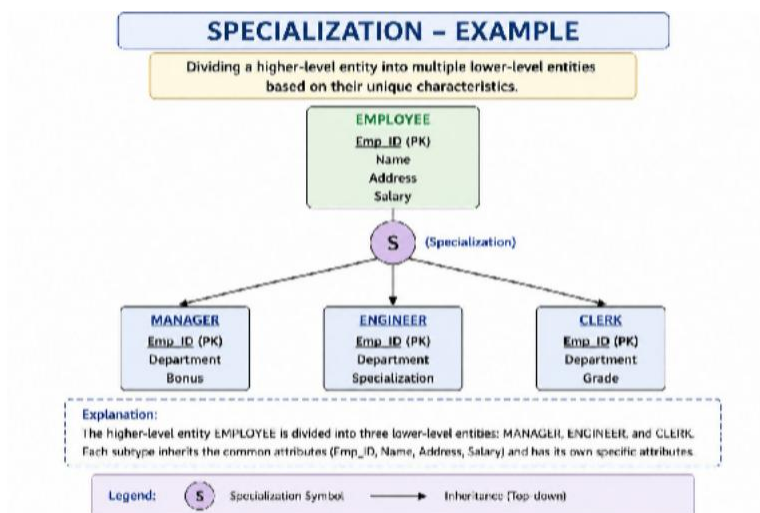


Both Savings and Current Accounts become one general entity called **Account**.

2. Specialization

Specialization divides one general entity into smaller specialized entities based on unique characteristics.

Example

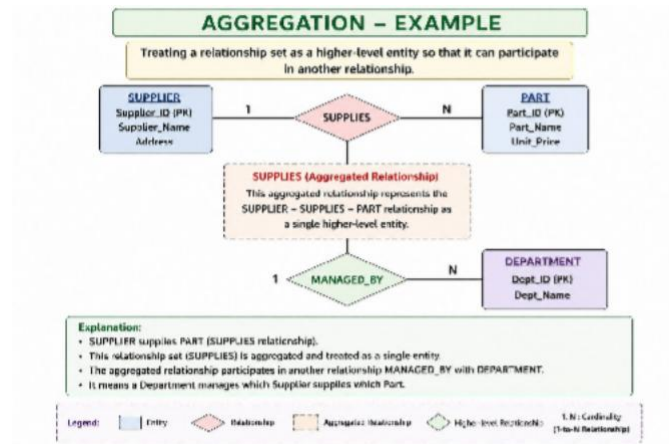


Employee is divided into Manager and Clerk.

3. Aggregation

Aggregation is used when a relationship itself participates in another relationship.

Example :



Here, the relationship "Treats" is treated as a single entity.

Advantages of EER Model

- Represents complex data easily.
- Reduces redundancy.
- Supports inheritance.
- Improves database design.
- Makes relationships more meaningful.

Conclusion

The EER model extends the ER model with advanced concepts such as generalization, specialization, and aggregation, enabling the design of more realistic and efficient databases.

Q4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.

Entities

Patient

- Patient_ID (PK)
- Name
- Age
- Gender

Doctor

- Doctor_ID (PK)
- Name
- Specialization

Department

- Department_ID (PK)
- Department_Name

Appointment

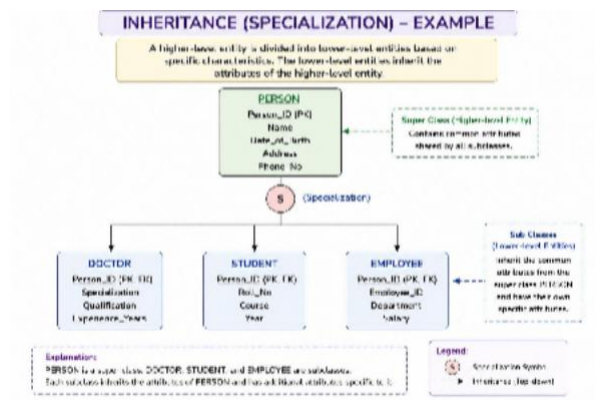
- Appointment_ID (PK)
- Date
- Time

Medical_Record (Weak Entity)

- Record_No (Partial Key)
- Diagnosis
- Treatment

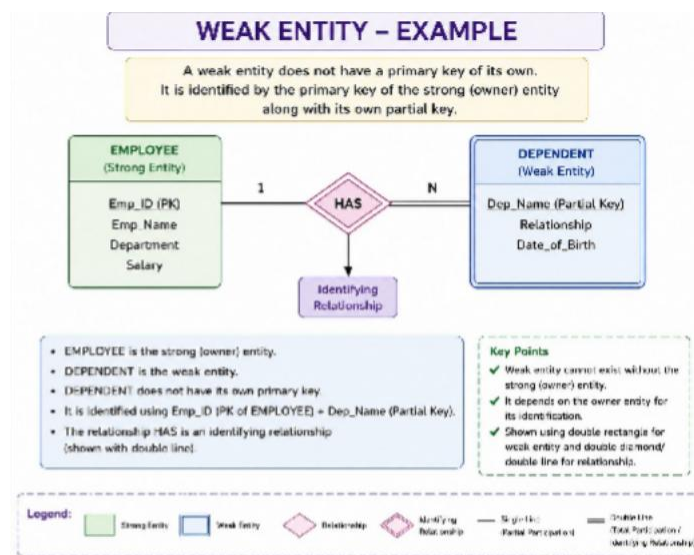
Medical_Record depends on Patient for its existence.

Inheritance (Specialization)

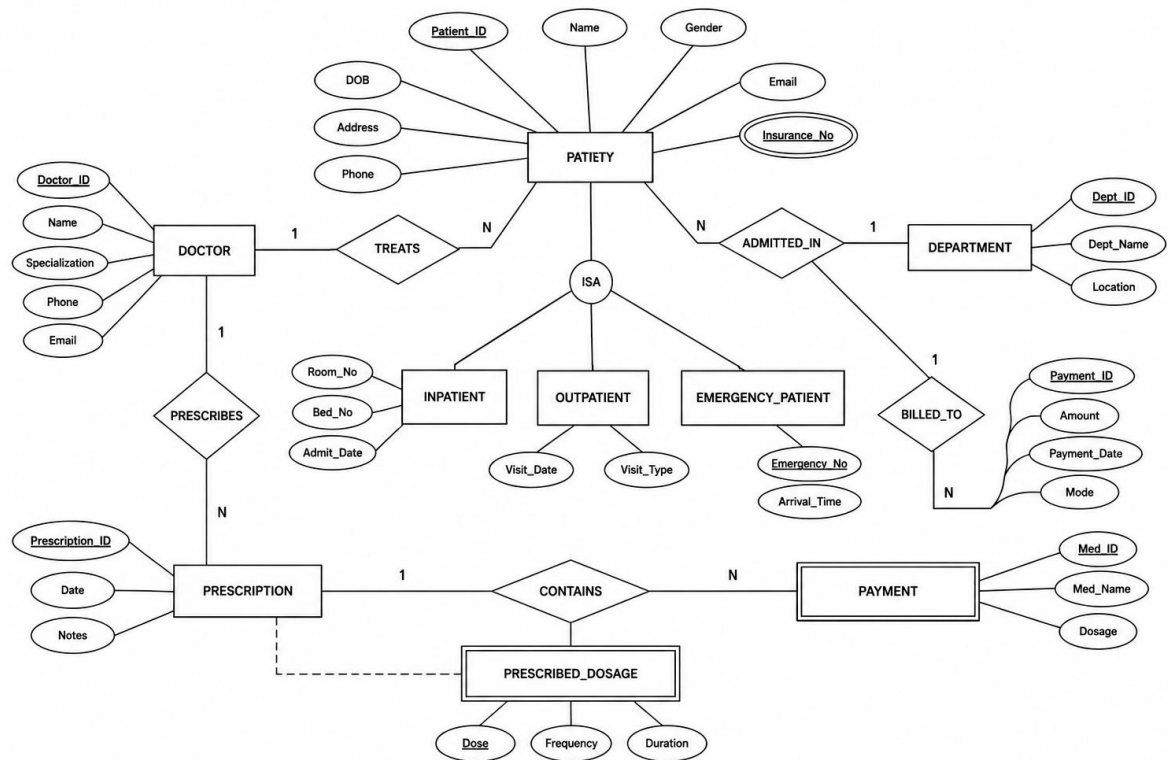


Doctor and Patient inherit common attributes from Person.

Weak Entity



Hospital EER Diagram :



Conclusion

The Hospital EER diagram uses inheritance through the Person entity and represents Medical_Record as a weak entity dependent on Patient, making the design more structured and efficient.

Q5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Database Administrator (DBA)

A **Database Administrator (DBA)** is a person responsible for installing, maintaining, securing, and managing a database. The DBA ensures that the database is available, reliable, and performs efficiently.

Responsibilities of a DBA:

1. Database Installation

Installs and configures the DBMS software.

2. Database Design

Creates database structures, tables, and relationships.

3. Security Management

Creates user accounts, grants permissions, and protects data from unauthorized access.

4. Backup and Recovery

Performs regular backups and restores data after failures.

5. Performance Monitoring

Optimizes queries, indexes, and storage to improve database performance.

6. Data Integrity

Ensures that stored data is accurate and consistent.

7. User Management

Creates users, assigns privileges, and manages roles.

8. Troubleshooting

Identifies and resolves database errors and failures.

9. Database Maintenance

Updates software, applies patches, and monitors storage.

10. Disaster Recovery

Prepares recovery plans to protect data during system failures.

Conclusion

A DBA plays a vital role in managing databases by ensuring security, performance, backup, recovery, and data integrity. The DBA helps organizations maintain reliable and efficient database systems